In [3]:
```
import pandas as pd
dataset=pd.read_csv("Social_Network.csv")
dataset
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [4]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
dataset
```

Out[4]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
...	...	...	...	...	...
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

400 rows × 5 columns

In [5]:
```
dataset=dataset.drop(columns=["User ID"])
dataset
```

Out[5]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [6]:
```
x=dataset[["Age","EstimatedSalary","Gender_Male"]]
y=dataset[["Purchased"]]
y
```

Out[6]:

	Purchased
0	0
1	0
2	0
3	0
4	0
...	...
395	1
396	1
397	1
398	0
399	1

400 rows × 1 columns

In [7]:
```
dataset.shape
```

Out[7]:

```
(400, 4)
```

In [8]:
```
dataset["Purchased"].value_counts()
```

Out[8]:

```
Purchased
0    257
1    143
Name: count, dtype: int64
```

In [9]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1/3,random_state=0)
```

In [18]:
```
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV
p_g={
    'gamma':['auto']
}
clf=GridSearchCV(SVC(),p_g,refit=True,verbose=3,n_jobs=-1)
clf.fit(x_train,y_train)
```

```
Fitting 5 folds for each of 1 candidates, totalling 5 fits
```

```
C:\Users\User\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1365: DataConversionWarning:
A column-vector y was passed when a 1d array was expected. Please change the shape of y to
(n_samples, ), for example using ravel().
  y = column_or_1d(y, warn=True)
```

Out[18]:

▸ **GridSearchCV** ⓘ ?

▸ **best_estimator_: SVC**

▸ SVC ?

In [19]:
```
y_pred=clf.predict(x_test)
y_pred
```

Out[19]:

```
array([0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
       0, 0])
```

In [20]:
```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_pred,y_test)
print(cm)
```

```
[[84 42]
 [ 1  7]]
```

In [21]:
```
from sklearn.metrics import classification_report
report=classification_report(y_pred,y_test)
print(report)
```

```
              precision    recall  f1-score   support

           0       0.99      0.67      0.80       126
           1       0.14      0.88      0.25         8

    accuracy                           0.68       134
   macro avg       0.57      0.77      0.52       134
weighted avg       0.94      0.68      0.76       134
```

In [ ]: